GAMEZONE SALES BEHAVIOR ANALYSIS

1. Project Overview

This project analyzes **GameZone order data** to understand customer purchasing patterns, product performance, and sales trends. The objective is to identify **top-selling games, customer purchase behavior, revenue trends, and order patterns** to support better business decisions and marketing strategies.

The dataset was analyzed using **Excel for data preparation and Power BI for visualization and dashboard creation**.

2. Dataset Summary

The **GameZone Orders dataset** contains transactional data related to online game purchases, including information about customers, products, marketing channels, and order fulfillment.

Dataset Characteristics

- **Total Rows:** 19,743
- **Total Columns:** 20

Missing Data – 208

Observations

- The **region column has the largest number of missing values**.
- Marketing and account creation columns have **78 missing values each**.
- Purchase-related columns have **minimal missing data**.

3. Data Analysis Using Excel

1. Total Revenue by Product

This analysis identifies which **games generate the highest revenue** based on product price and purchase volume.

This helps determine **which games contribute the most to overall revenue**.

Row Labels	Sum of usd_price_cleaned
27in 4K gaming monitor	$ 1,742,238.49
Acer Nitro V Gaming Laptop	$ 57,456.44
Dell Gaming Mouse	$ 32,754.03
JBL Quantum 100 Gaming Headset	$ 89,140.59
Lenovo IdeaPad Gaming 3	$ 670,780.20
Nintendo Switch	$ 1,504,363.78
Razer Pro Gaming Headset	$ 884.23
Sony PlayStation 5 Bundle	$ 1,393,873.98
Grand Total	**$ 5,491,491.74**

2. Sales by Region

This analysis determines **which regions generate the highest revenue**.

The insights help guide **regional marketing strategies and expansion opportunities**.

Row Labels	Sum of usd_price_cleaned	
APAC	$	645,869.71
EMEA	$	1,658,960.10
LATAM	$	306,950.35
NA	$	2,669,877.96
North America	$	206,702.51
Grand Total	**$**	**5,488,360.63**

4. Revenue by Platform

Revenue is compared across different **purchase platforms**, such as:

- Website
- Mobile

This helps determine **which platform customers prefer when purchasing games**.

Row Labels	Sum of usd_price_cleaned	
mobile app	$	141,733.81
website	$	5,349,757.93
Grand Total	**$**	**5,491,491.74**

5. Marketing Channel Performance

This analysis identifies which **marketing channels drive the most purchases**, such as:

Row Labels	Sum of usd_price_cleaned
affiliate	$ 196,338.82
direct	$ 4,642,799.26
email	$ 554,002.06
social media	$ 65,563.77
Grand Total	**$ 5,458,703.91**

5. Monthly Sales Trends

Sales were analyzed over time to understand **purchase trends across months and years**.

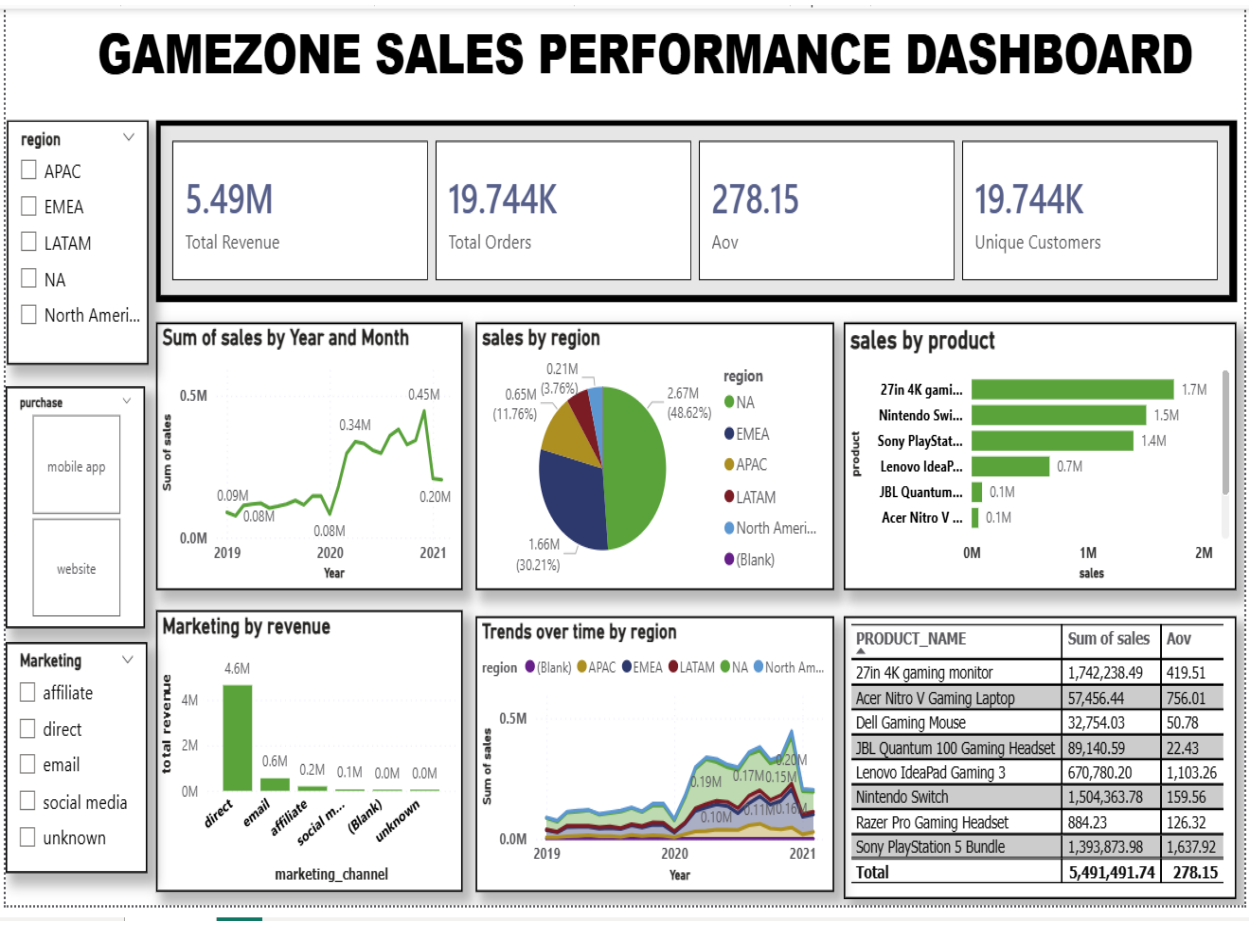
This helps identify:

- seasonal purchasing behavior
- peak sales periods

Sum of usd_price_cleaned	Column Labels								
Row Labels	27in 4K gaming monitor	Acer Nitro V Gaming Laptop	Dell Gaming Mouse	JBL Quantum 100 Gaming Headset	Lenovo IdeaPad Gaming 3	Nintendo Switch	Razer Pro Gaming Headset	Sony PlayStation 5 Bundle	Grand Total
1/1/1900				$ 21.96					21.96
1/1/2019	$ 29,531.00	$ 1,500.21		$ 2,525.90	$ 6,136.29	$ 34,296.72		$ 14,933.71	88923.83
2/1/2019	$ 24,590.88	$ 1,463.76		$ 1,786.18	$ 7,548.12	$ 24,877.51		$ 16,042.02	76308.47
3/1/2019	$ 34,262.28	$ 3,032.40		$ 3,432.28	$ 14,864.02	$ 37,742.99		$ 19,589.03	112923
4/1/2019	$ 46,298.00	$ 2,126.96		$ 3,304.32	$ 10,771.68	$ 36,282.65		$ 18,885.26	117668.87
5/1/2019	$ 37,622.04			$ 3,096.32	$ 12,594.76	$ 36,485.51		$ 31,270.71	121069.34
6/1/2019	$ 35,789.14	$ 1,581.28		$ 3,058.05	$ 13,328.81	$ 33,240.05		$ 16,749.25	103746.58
7/1/2019	$ 41,345.38	$ 739.60		$ 3,087.60	$ 14,018.67	$ 36,100.74		$ 14,871.45	110163.44
8/1/2019	$ 38,094.84	$ 648.22		$ 2,836.54	$ 13,986.91	$ 41,800.85		$ 20,460.10	117827.46
9/1/2019	$ 49,365.33	$ 798.00		$ 3,293.54	$ 14,192.79	$ 42,437.46		$ 20,507.26	130594.38
10/1/2019	$ 45,705.59	$ 1,574.26		$ 2,636.95	$ 10,980.86	$ 34,864.78		$ 19,226.23	114988.67
11/1/2019	$ 57,687.95	$ 1,361.44		$ 2,880.71	$ 19,938.08	$ 46,673.79		$ 17,812.39	146354.36
12/1/2019	$ 54,556.90	$ 745.04		$ 3,602.30	$ 15,837.13	$ 52,465.77		$ 19,192.22	146399.36
1/1/2020	$ 24,025.01		$ 49.98	$ 1,428.92	$ 10,425.92	$ 22,442.38		$ 24,053.21	82425.42
2/1/2020	$ 60,064.64	$ 2,394.00	$ 202.58	$ 2,931.36	$ 21,339.53	$ 47,124.62		$ 46,756.07	180812.8
3/1/2020	$ 96,366.17	$ 2,249.81	$ 485.53	$ 4,246.14	$ 40,423.75	$ 69,341.45	$ 119.98	$ 83,596.94	296829.77
4/1/2020	$ 101,988.77		$ 1,269.87	$ 5,014.58	$ 33,848.14	$ 86,161.52	$ 112.83	$ 109,830.58	338226.29
5/1/2020	$ 108,407.82	$ 3,830.40	$ 1,261.40	$ 5,096.63	$ 38,958.11	$ 91,827.93	$ 119.98	$ 81,939.62	331441.89
6/1/2020	$ 94,248.44	$ 1,109.64	$ 1,383.40	$ 4,309.48	$ 32,101.63	$ 90,566.05	$ 119.98	$ 84,423.28	308261.9
7/1/2020	$ 90,498.92	$ 2,394.00	$ 1,435.04	$ 4,224.13	$ 32,898.30	$ 86,799.11		$ 78,843.23	297092.73
8/1/2020	$ 87,045.85	$ 4,060.30	$ 1,775.11	$ 3,793.04	$ 59,236.71	$ 93,137.70	$ 138.26	$ 109,955.61	359142.58
9/1/2020	$ 112,800.83	$ 5,540.18	$ 4,164.15	$ 4,601.44	$ 54,047.91	$ 95,437.72		$ 104,597.17	381189.4
10/1/2020	$ 103,411.72	$ 2,222.03	$ 3,965.78	$ 3,543.63	$ 34,288.21	$ 70,615.92	$ 119.98	$ 109,548.77	327716.04
11/1/2020	$ 106,857.85	$ 3,938.70	$ 4,848.80	$ 4,385.00	$ 36,631.91	$ 83,671.09	$ 153.22	$ 102,436.82	342923.39
12/1/2020	$ 126,740.31	$ 8,477.01	$ 4,991.32	$ 5,006.94	$ 68,405.17	$ 100,955.12		$ 131,735.29	446311.16
1/1/2021	$ 63,219.91	$ 1,596.00	$ 3,707.40	$ 2,226.15	$ 23,583.20	$ 57,642.73		$ 56,127.50	208102.89
2/1/2021	$ 71,712.92	$ 4,073.20	$ 3,213.67	$ 2,770.50	$ 30,393.59	$ 51,371.62		$ 40,490.26	204025.76
Grand Total	**$ 1,742,238.49**	**$ 57,456.44**	**$ 32,754.03**	**$ 89,140.59**	**$ 670,780.20**	**$ 1,504,363.78**	**$ 884.23**	**$ 1,393,873.98**	**5491491.7**

4. Power BI Dashboard

An interactive **Power BI dashboard** was developed to present insights visually.



5. Business Recommendations

Based on the analysis, the following recommendations are suggested.

1. Promote Top-Revenue Games

Highlight the highest-revenue games in marketing campaigns.

2. Improve Marketing Channel Strategy

Focus investment on **marketing channels generating the most conversions**.

3. Target High-Revenue Regions

Focus advertising campaigns on **regions that generate the highest sales**.

4. Optimize Platform Experience

Improve the purchasing experience on the **most frequently used platform**.

5. Plan Seasonal Promotions

Use sales trend data to launch **promotions during peak purchasing periods**.